

An ideal WYSIWYG editor over a UI tree, and its non-removable restriction

Abstract

We model an editor over the UI tree $T = (N, r, p, \chi, \ell)$ as a labelled transition system on *workspaces*. The “ideal” editor—one in which *anything* well-formed can be built—and a “restricted” editor—in which a fixed handful of elements may be styled, moved, and otherwise manipulated but never removed—are not different mechanisms. They are the same construction at two ends of a lattice of invariants: the ideal editor is the top element, constrained only by well-formedness, and a restricted editor is the sub-transition-system carved out by a stronger invariant.

1 The state space

A *workspace* is a tuple $W = (N, p, \chi, \ell, r)$ in which (N, p) is a finite forest, $r \in N$ a distinguished root, χ orders children, and ℓ labels nodes by elements $\delta = (\tau, A, S, E, \rho)$. The *document* is the part reachable from r ,

$$\text{Doc}(W) := \{ n \in N : r \triangleleft^* n \},$$

and the remaining forest-roots head *detached fragments*, living in $N \setminus \text{Doc}(W)$ (the c_0, c_1, c_2 of the bracket figure). We write $\mathcal{T}$ for the set of *well-formed* workspaces, those satisfying:

WF1. $p : N \rightarrow N$ is a partial function: every node has at most one parent.

WF2. $\chi(n)$ is a duplicate-free sequence enumerating exactly $p^{-1}(n)$; thus χ is p together with a sibling order.

WF3. $\triangleleft^+$ is irreflexive: the tree is acyclic and following p terminates. With WF1 this makes (N, p) a forest.

WF4. $r \notin \text{dom}(p)$ and $\text{Doc}(W)$ is the r -reachable set (the root-reachability condition of the bracket figure).

WF5. ℓ is total, and each $\delta = (\tau, A, S, E, \rho)$ is well-typed over a fixed tag alphabet.

Remark 1.1 (Schema). One may impose a sixth, optional constraint: a *schema* Σ , a content model fixing which types τ may parent which. This is the hinge of what follows. “Anything can be created” is precisely the statement that Σ is trivial; the ideal editor is not powerful, it merely imposes nothing past WF1–WF5. Any genuine schema is already a restriction of the kind developed in Section 3.

2 The ideal editor

An *editor* is a labelled transition system $\mathcal{E} = (\mathcal{S}, \mathcal{O}, \rightarrow)$ with $\mathcal{S} \subseteq \mathcal{T}$ and each operation a *partial function on workspaces*, defined only when its result again lies in $\mathcal{T}$. The following five operations generate everything.

- (1) $\text{mint}(\delta) \Rightarrow n$: choose a fresh $n \notin N$, set $\ell(n) = \delta$, and leave n parentless—a singleton fragment. *Always defined.*
- (2) $\text{attach}(n, q, k)$: pre. n is a fragment root, $q \in N$ lies outside the subtree of n (no cycle), and $0 \leq k \leq |\chi(q)|$. Sets $p(n) = q$ and inserts n at index k of $\chi(q)$.
- (3) $\text{detach}(n)$: pre. $n \in \text{dom}(p)$ and $n \neq r$. Removes n from $\chi(p(n))$ and unsets $p(n)$; the subtree of n leaves with it. *Orphan, not destroy.*
- (4) $\text{delete}(n)$: pre. n is a fragment root and $n \neq r$. Removes n (with its subtree) from N . *Garbage-collect.*
- (5) $\text{relabel}(n, \delta')$: sets $\ell(n) = \delta'$; refine into $\text{setType}, \text{setAttr}, \text{setStyle}, \dots$ to touch a single component of δ .

Write $\text{move}(n, q, k) := \text{detach}(n); \text{attach}(n, q, k)$ for the derived reparent/reorder (with $q = p(n)$ a pure reorder), and call the generating set

$$\mathcal{O}_{\top} = \{\text{mint}, \text{attach}, \text{detach}, \text{delete}, \text{relabel}\}.$$

The separation of **detach** (remove from view) from **delete** (destroy) is what makes the restriction of Section 3 clean.

Theorem 2.1 (Universality). *From the seed $W_{\emptyset}$ (a single root r), the set reachable under $\mathcal{O}_{\top}$ is exactly $\mathcal{T}$, and the reachability graph is strongly connected.*

Proof sketch. Soundness. Each precondition forces the result back into WF1–WF5: **mint** adds an isolated labelled node; **attach**, by its no-cycle precondition, keeps the forest and updates χ consistently; **detach** splits off a subtree; **delete** drops a whole tree; **relabel** touches only ℓ .

Completeness. To realise a target W^* , mint every node with its target label, then visit the nodes in *preorder* and **attach** each non-root beneath its target parent—preorder guarantees the parent is already present—inserting at the target index. Detached fragments are rebuilt the same way.

Strong connectivity. From any W , repeatedly **detach** a leaf and **delete** it, reaching $W_{\emptyset}$; composing with the build sequence reaches any W' . $\square$

Thus “an ideal WYSIWYG editor such that anything can be created” unfolds into *soundness* (every operation preserves $\mathcal{T}$) together with *completeness* (reachable = $\mathcal{T}$), under the trivial schema of Remark 1.1.

3 The non-removable restriction

Fix a finite set of *protected identities* $P \subseteq \mathbb{I}$ —the small, manipulable-but-permanent set—optionally with frozen types $\theta: P \rightarrow \text{Tag}$. The entire restriction is one invariant on the *successor* state.

Definition 3.1 (Protected invariant and restricted steps).

$$\begin{aligned}\Phi_P^\theta(W) &: \iff P \subseteq \text{Doc}(W) \wedge \forall p \in P. \tau(\ell(p)) = \theta(p), \\ W \xrightarrow[P]{e} W' &: \iff (W \xrightarrow{e} W') \wedge \Phi_P^\theta(W').\end{aligned}$$

That single condition needs no separate per-operation blocklist; reading it off each operation yields Table 1.

operation	effect on a P -node	outcome
delete	drops it from $N \supseteq \text{Doc}$	blocked (cannot remove)
detach	moves it out of Doc	blocked (cannot orphan)
detach of an <i>ancestor</i>	carries a P -node out of Doc	blocked
move to an in-document parent	subtree stays in Doc	allowed (“moved”)
setStyle, setAttr	membership and type untouched	allowed (“styled”)
setType to $\tau' \neq \theta(p)$	breaks the type clause	blocked (role frozen)
any op on a non- P node, mint	cannot threaten Φ	allowed (free)

Table 1: Permissions induced by Φ_P^θ .

The third row is the non-obvious consequence worth naming.

Proposition 3.2 (Removability propagates upward). *Under $\xrightarrow{P}$ the nodes that can no longer be detached or deleted are exactly the ancestor-closure of P ,*

$$\text{anc}^*(P) = \{m \in N : \exists p \in P. m \triangleleft^* p\},$$

since orphaning any such m would carry a protected node out of the document. Protecting a node therefore freezes the removability of its entire ancestor chain, while leaving each protected node free to move downward or sideways wherever WF permits.

Theorem 3.3 (Restricted reachability). *From any seed W_0 with $\Phi_P^\theta(W_0)$ (the protected identities are given, not minted), the set reachable under $\xrightarrow{P}$ is exactly*

$$\{W \in \mathcal{T} : \Phi_P^\theta(W)\}.$$

It is strongly connected within this invariant but can never leave it. Formally this is the sub-transition-system induced by Φ_P^θ : keep only states satisfying the invariant and only edges between them. That induced subgraph is the precise meaning of “a restricted subset of the ideal editor.”

Proof sketch. The build/teardown argument of Theorem 2.1 goes through verbatim on the non-protected part; protected nodes are never deleted, detached, or retyped, and are repositioned only by **move**, which preserves Φ_P^θ by Table 1. Conversely any target satisfying the invariant is reached by editing the non-protected structure and **move**-ing the protected nodes into place. $\square$

For finer, per-element control, replace the global P by a *capability map*

$$\kappa : N \longrightarrow 2^{\{\text{remove, reparent, reorder, restyle, retype, addChild, rmChild}\}},$$

permitting an operation of kind k on node n iff $k \in \kappa(n)$. The protected set is the instance $\kappa(p) = \{\text{reparent, reorder, restyle}\}$ for $p \in P$ and $\kappa(n) = \text{ALL}$ otherwise.

Remark 3.4 (Carrying policy in the element). The natural home for κ is the element itself. If the fifth component ρ of $\delta = (\tau, A, S, E, \rho)$ carries (or is extended to carry) per-node policy, the restricted editor is *literally* the ideal editor reading editability off each node’s ρ , and “locked” status travels with the node across moves.

4 The WYSIWYG qualifier

Equip the document with a *render* $R: \mathcal{T} \rightarrow \mathcal{V}$ (layout and paint of $\text{Doc}(W)$, ignoring fragments). The user acts on the *view* by gestures g ; a WYSIWYG editor supplies a *lift* L sending (W, g) to an operation so that editing the rendered output *is* editing the state:

$$\begin{array}{ccc} W & \xrightarrow{L(W, g)} & W' = W \cdot L(W, g) \\ R \downarrow & & \downarrow R \\ R(W) & \xrightarrow{g} & R(W') \end{array}$$

$$R(W \cdot L(W, g)) = g \cdot R(W).$$

“WYSIWYG and anything-creatable” then asks that R be injective on editing-relevant structure—no hidden state alters the picture without a handle—and that L be total over the gestures needed to reach any view. At that point the universality of Theorem 2.1 transports across R to give *completeness of direct manipulation*: every renderable view is reachable by gesture.

The restriction of Section 3 surfaces honestly as *missing affordances*: for $p \in P$ the lift offers drag-to-restyle and drag-to-reposition handles but no delete handle, because the lifted operation would fail Φ_P^θ .

Remark 4.1 (Honest caveat). True injectivity of R is essentially always false—distinct states render to identical pixels—so production WYSIWYG editors take R against a structured *editing model* rather than the literal raster, and state the square there.

5 Composing restrictions

Schema and protection are invariants of the same lattice and simply intersect: the editor constrained by both is the sub-transition-system induced by $\Phi_P^\theta \wedge \Sigma$. The ideal editor is the top, $\Phi \equiv \top$; every editor one actually ships is some meet of invariants below it.

